

06

Using Google Cloud Managed Storage Services with GKE

Using Google Cloud Managed Storage Services with Google Kubernetes Engine

- 01 Managed and self-managed storage services
- 02 Cloud Storage use cases
- 03 Google Cloud managed database services
- 04 Cloud SQL and Cloud SQL Auth Proxy
- 05 Google Cloud storage options



Using Google Cloud Managed Storage Services with Google Kubernetes Engine

01 Managed and self-managed storage services

02 Cloud Storage use cases

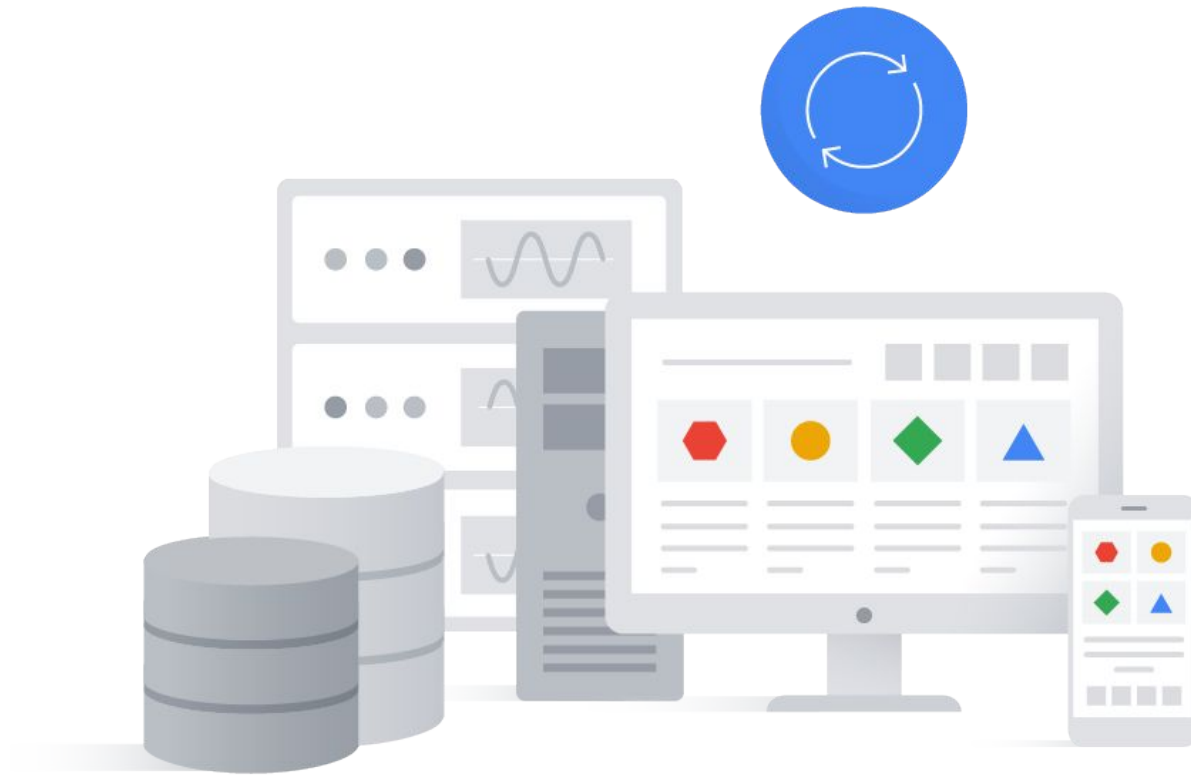
03 Google Cloud managed database services

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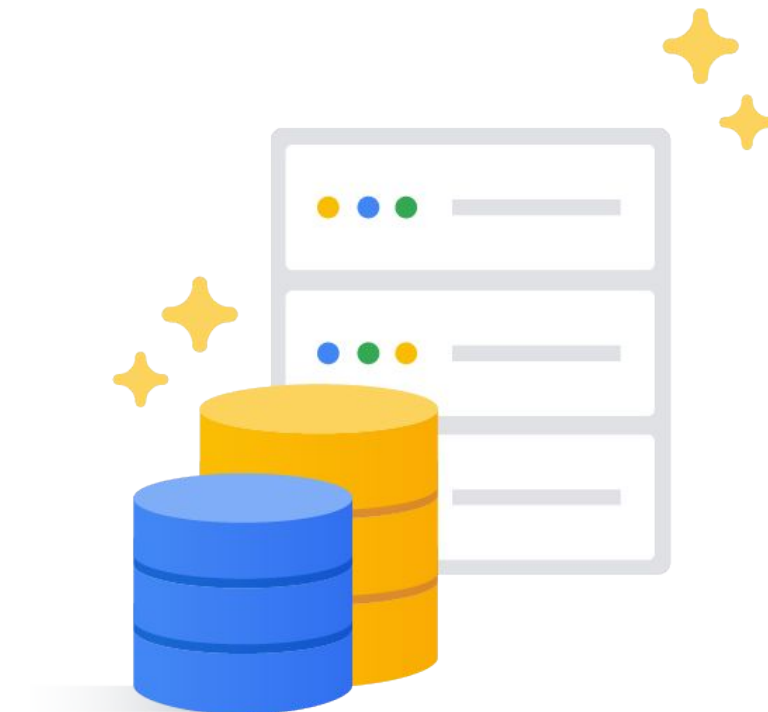
GKE supported storage options



Option 1: Store database files using a GKE Volume based on Persistent Disks.

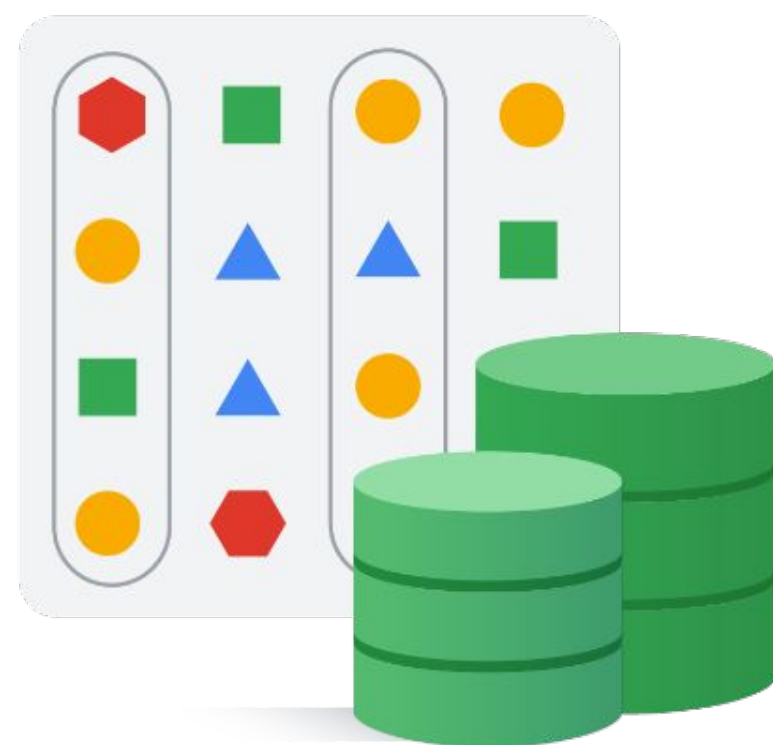


You are responsible for managing the lifecycle and service.

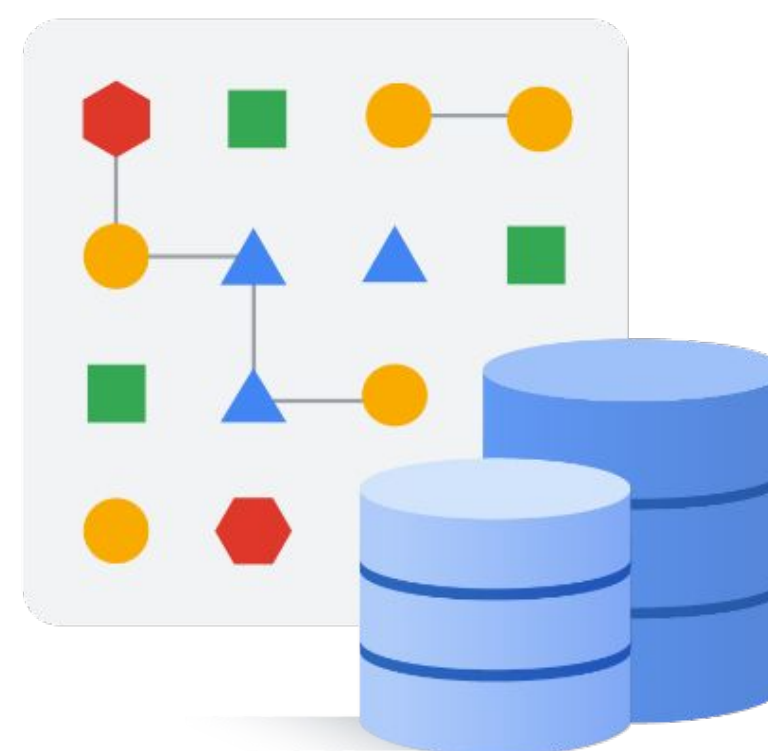


Option 2: Use Google Cloud's fully managed database and storage services.

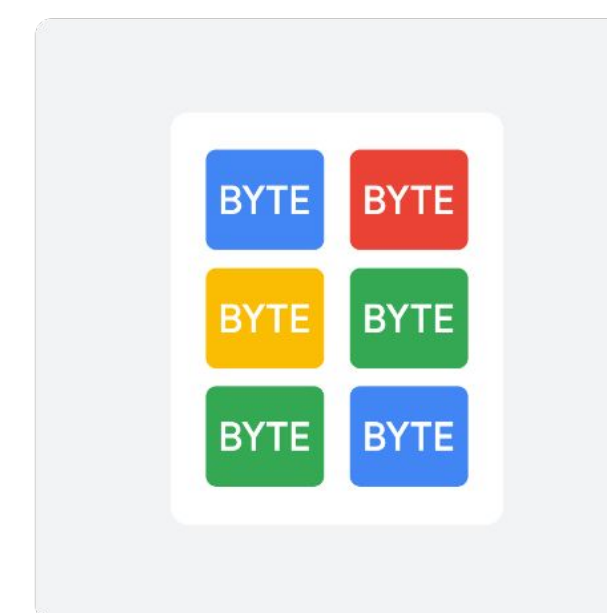
Fully managed database and storage services



Relational databases

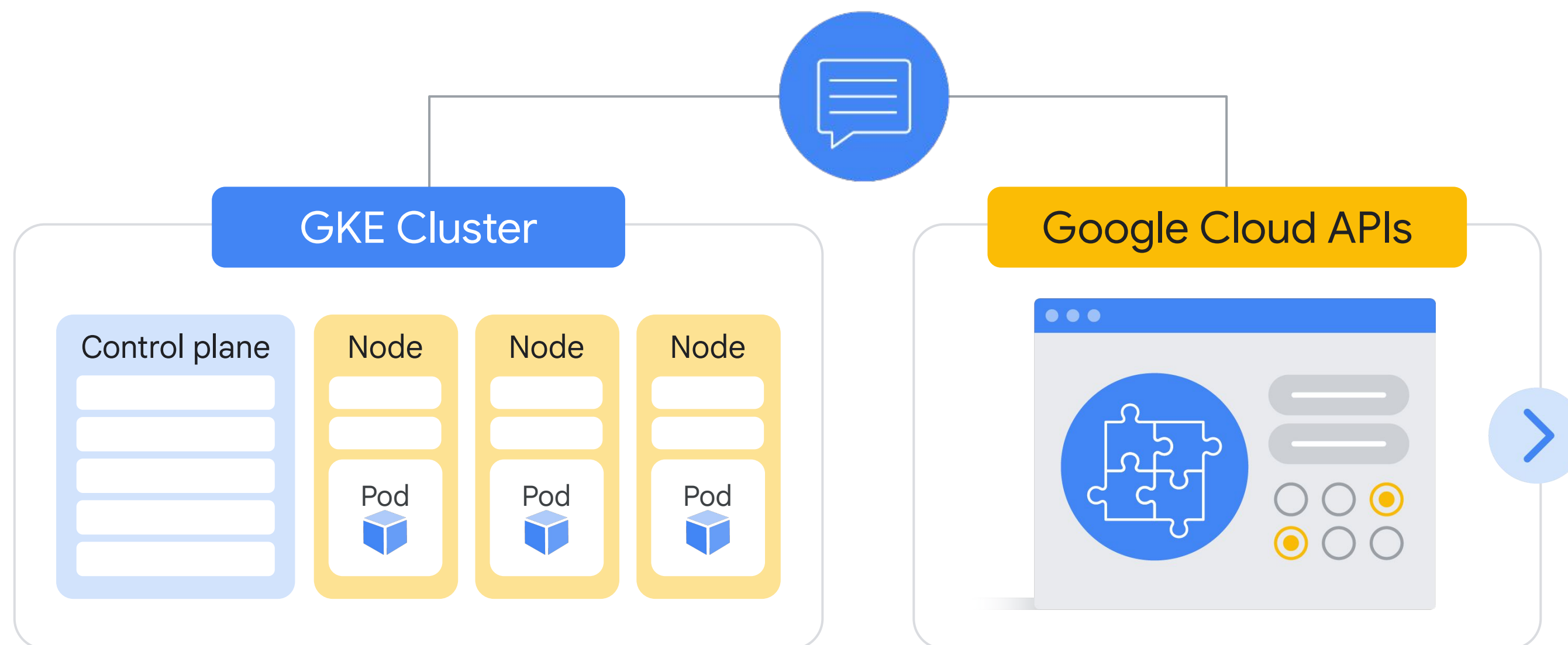


Non-relational databases



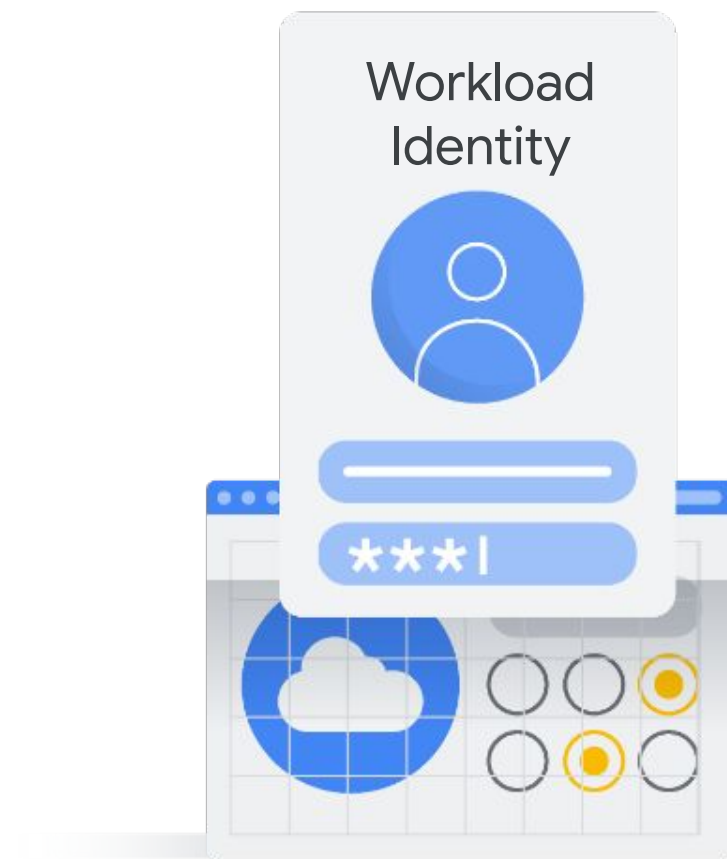
Object storage

Access managed services in GKE

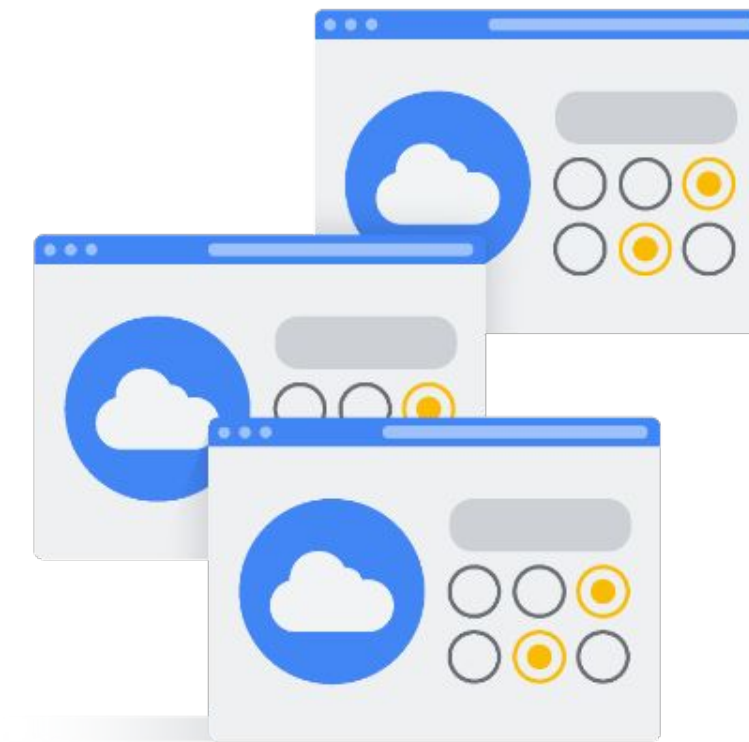


The app uses authorized credentials to perform tasks after the API is enabled.

Simplify process management with Workload Identity

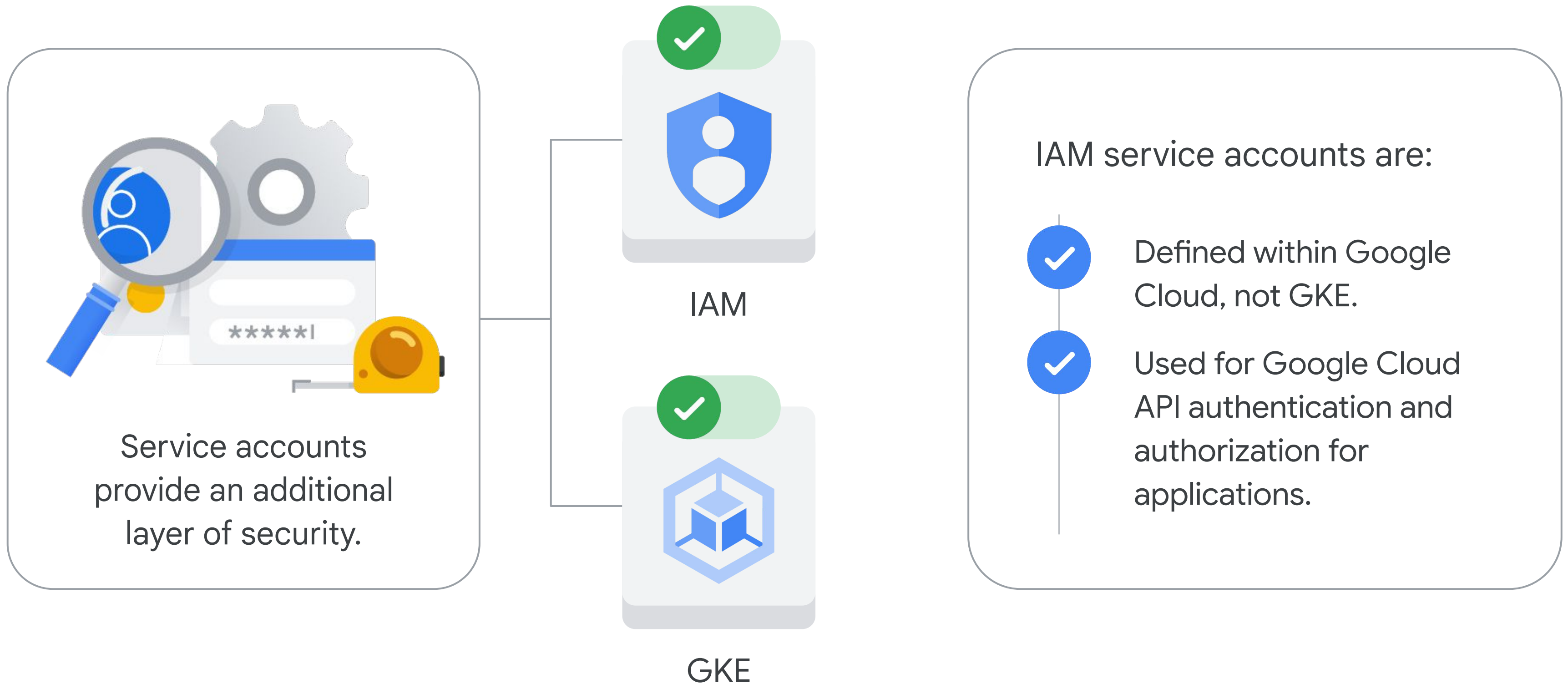


Uses dedicated credentials to directly authenticate.



Eliminates the need to manage a collection of service account keys for each Google Cloud API you want to use.

Service accounts



IAM policies can bind services accounts

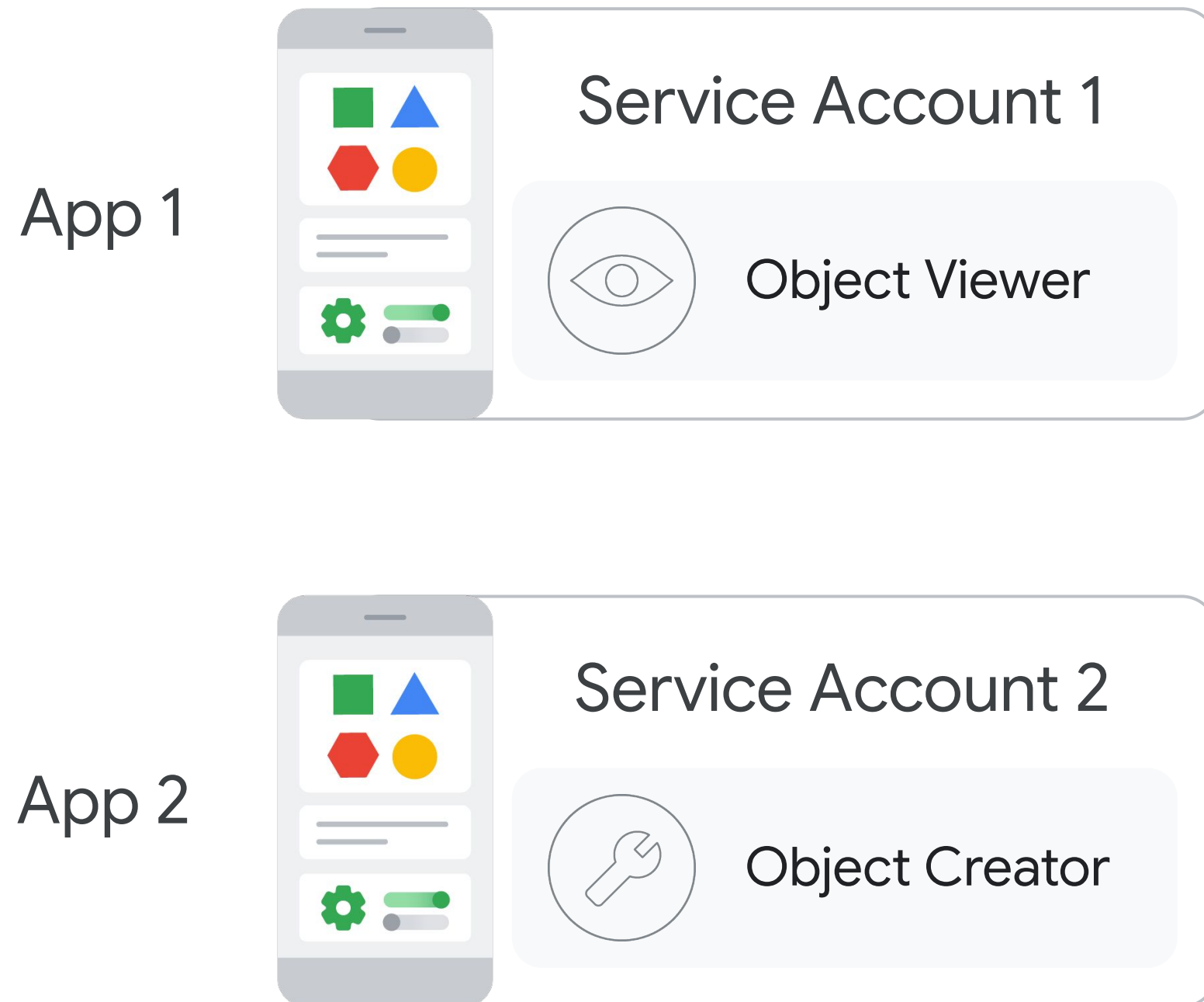
```
gcloud projects add-iam-policy-binding  
projects/PROJECT_ID \  
  
--role=roles/container.clusterViewer \  
  
--member=principal://iam.googleapis.com/  
projects/PROJECT_NUMBER/locations/global/  
workloadIdentityPools/PROJECT_ID.svc.id.goog/  
subject/ns/NAMESPACE/sa/KSA_NAME \  
  
--condition=None
```

An IAM policy can be used to bind an IAM service account to a GKE service account.

Annotate the Kubernetes ServiceAccount object with the name of the IAM service account, and assign it to the workloads that will use the Google Cloud APIs.

The IAM service account must have an IAM role that matches your application's required permissions.

It is a best practice to use separate service accounts



Create a dedicated service account for each application that interacts with Google Cloud services.

Grant each service account only the specific IAM roles and permissions it needs.

If an attacker notices many service accounts, the value of each will appear to be less valuable.

Using Google Cloud Managed Storage Services with Google Kubernetes Engine

01 Managed and self-managed storage services

02 Cloud Storage use cases

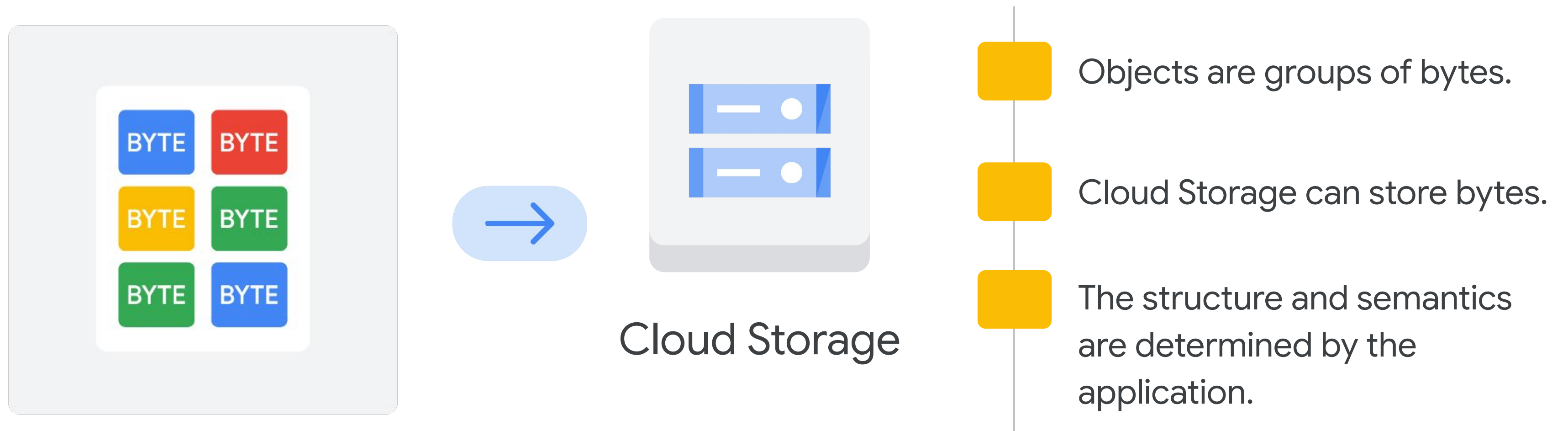
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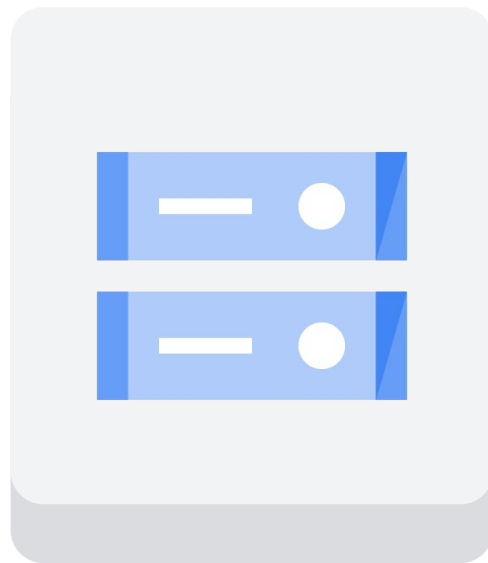
05 Google Cloud storage options



Cloud Storage is an object storage service



Cloud Storage



Cloud Storage



Commonly used for media hosting: serving images, streaming music and videos.



It can be used as a data lake for analytics and machine learning workloads.



It can store unstructured and structured files and data, but it's not optimized for querying.



It's not designed to be a file system, so it won't replace persistent volumes.

Cloud Storage's four primary storage classes

	Standard Storage	Best for frequently accessed data or data that's stored for only brief periods of time.
	Nearline Storage	Best for storing infrequently accessed data, like reading or modifying data on average once a month or less.
	Coldline Storage	A low-cost option for storing infrequently accessed data.
	Archive Storage	Lowest-cost option for data archiving, online backup, and disaster recovery.

Cloud Storage access control features



Encryption

Data is encrypted on the server-side with encryption keys.



Pub/Sub integration

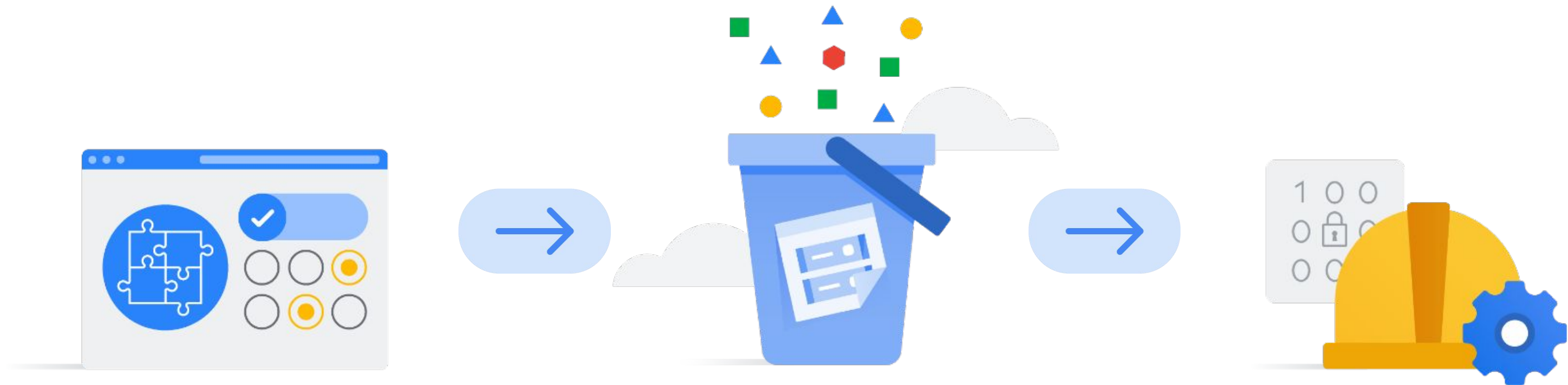
Notifies apps when changes are made to a Cloud Storage bucket.



Versioning

Versioning can be enabled for Cloud Storage.

Use APIs to access Cloud Storage



After Google Cloud APIs are turned on for your applications...

...they can be used to access Cloud Storage.

Important:

Your application will need to use a service account to access the service you want to use.

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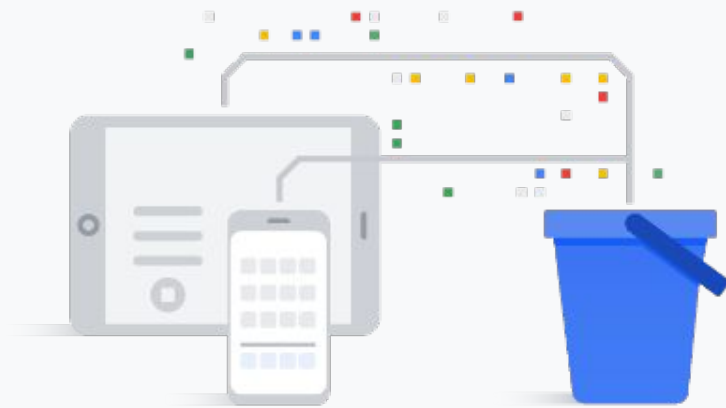
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Google Cloud's database purposes



01

Support online
application data
storage and retrieval

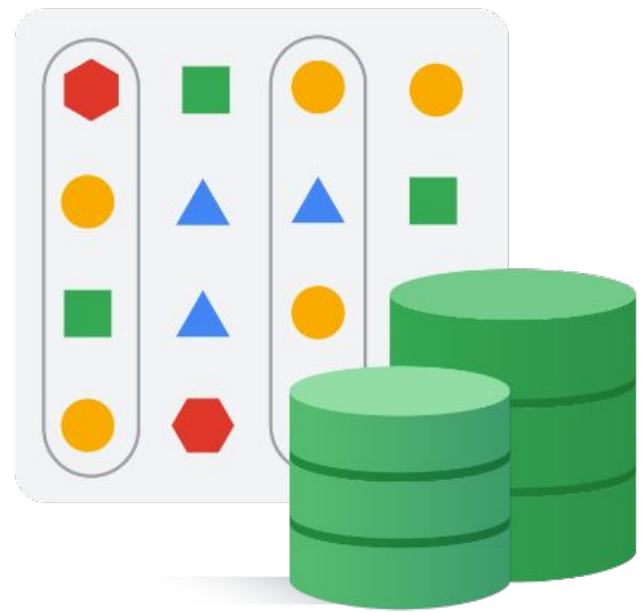


02

Provide an analytics
database optimized for
data mining and trend
analysis

Google Cloud's database options

Relational databases



- Store and provide data points that are related to one another.
- Include built-in mechanisms for consistency and integrity.

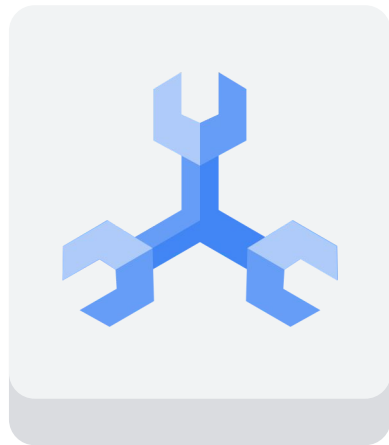
Non-relational databases



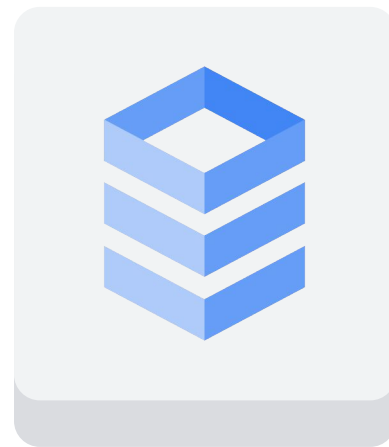
- Less structured in format than relational databases.
- Prioritize flexibility over built-in data integrity.

Google Cloud database products

Relational databases

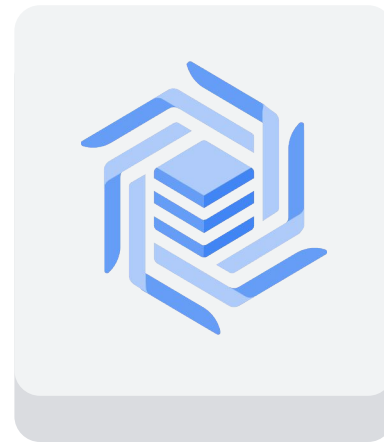


Spanner

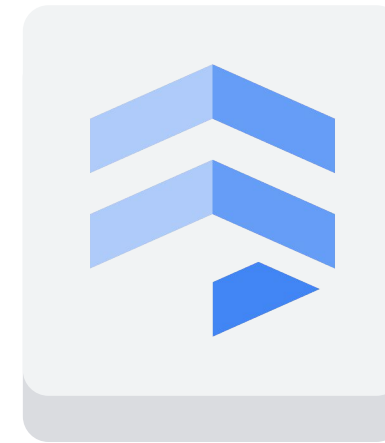


Cloud SQL

Non-relational databases

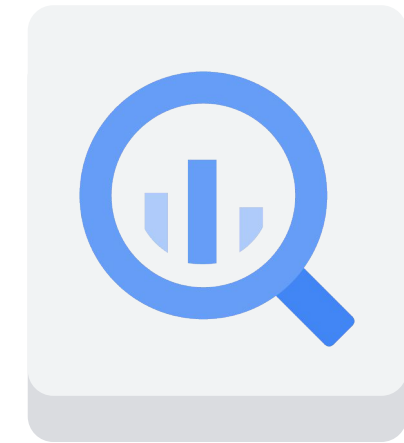


Bigtable



Firestore

Data warehouse



BigQuery

Bigtable



Bigtable

Google's NoSQL big data database service

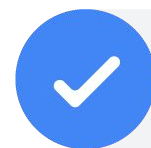


Designed to handle large workloads at consistent low latency

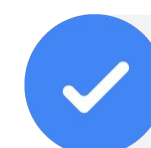


Can send and receive large amounts of data

Bigtable is a great choice for:

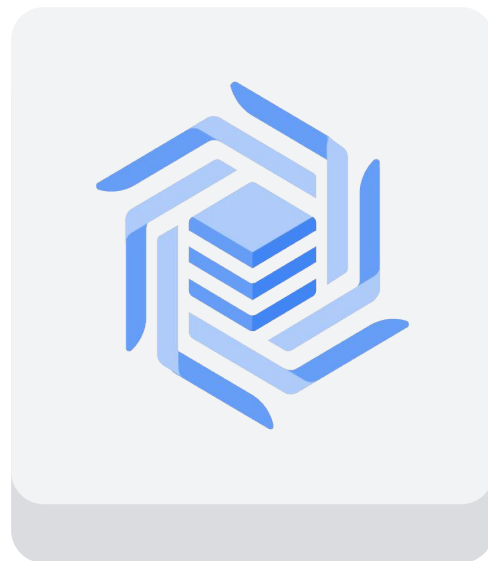


Operational applications



Analytical applications

Consider Bigtable for:



Bigtable



Storing and accessing more than one terabyte of semi-structured or structured data.



High throughput or rapidly changing data.



Data transactions where strong relational semantics are not required.



Time-series data or data with natural semantic ordering.

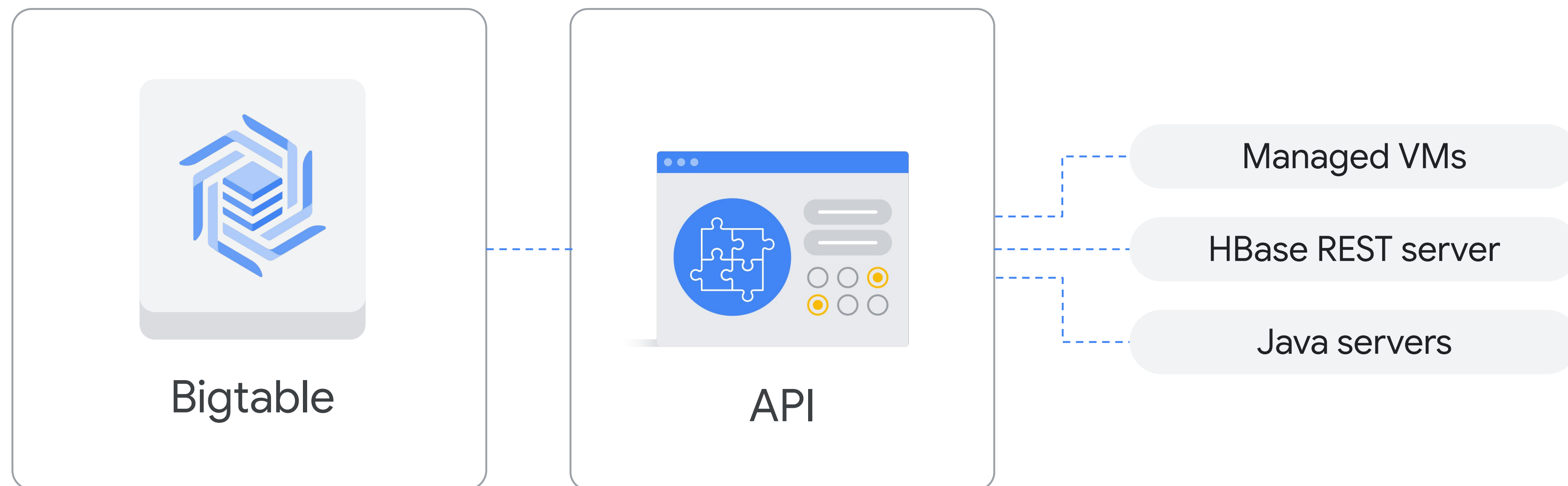


Running asynchronous batch or synchronous real-time processing on big data.



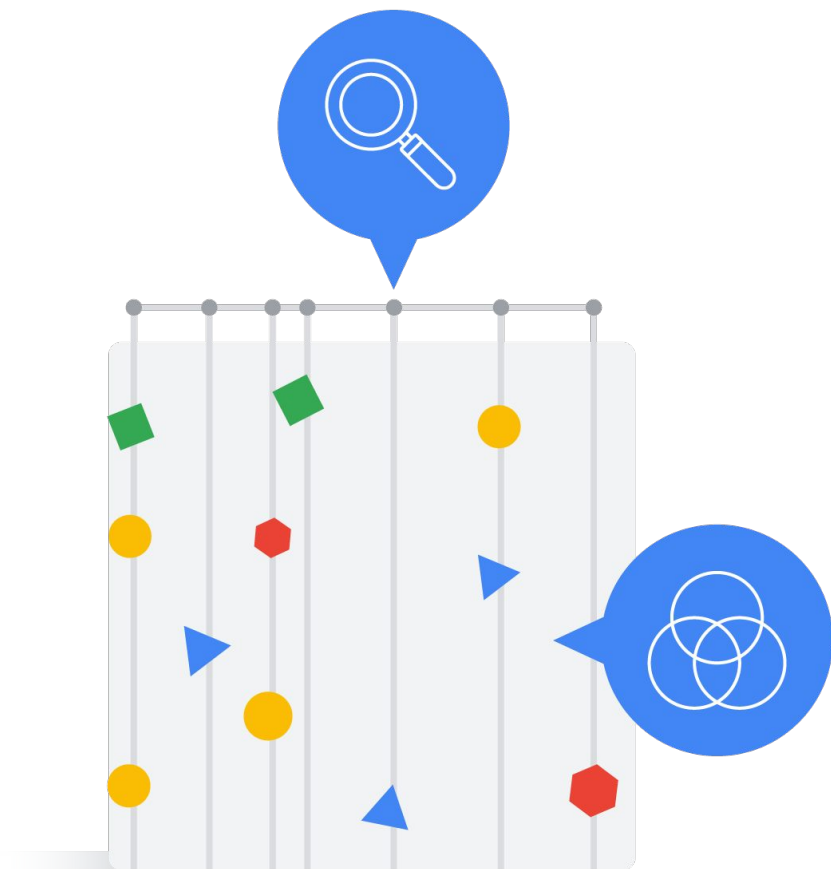
Running machine learning algorithms on the data.

APIs and Bigtable



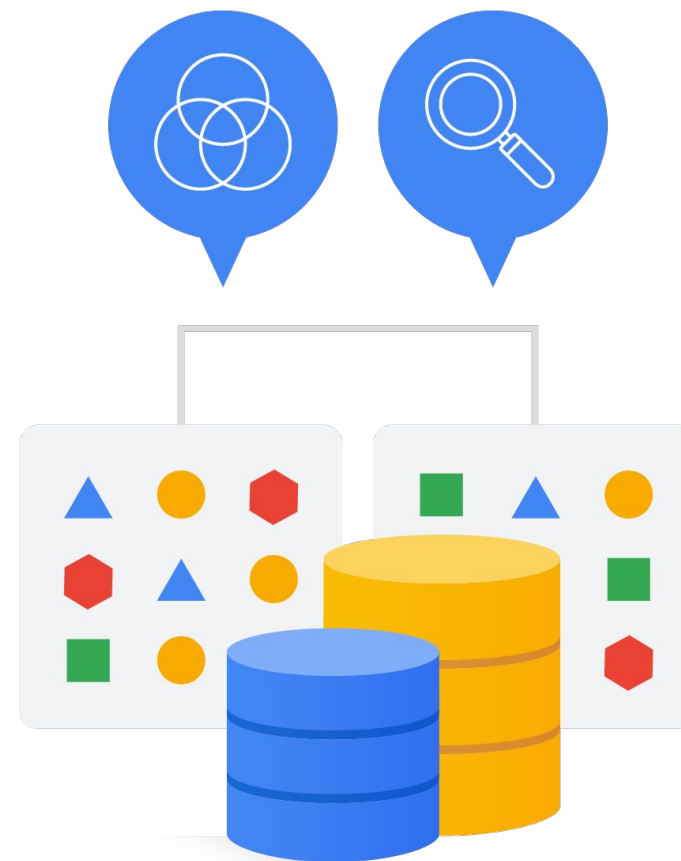
Bigtable supports streaming and batch data

Streaming data



Dataflow Streaming Engine,
Spark Streaming, or Storm

Batch data



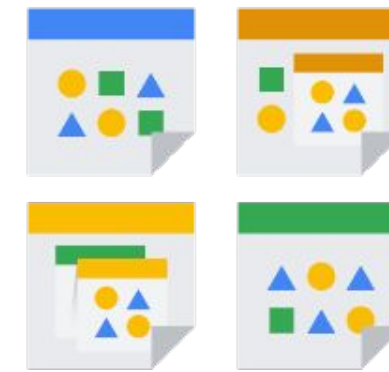
Hadoop MapReduce,
Dataflow, and Spark

Often processed data
is written back to
Bigtable or another
database for further use.

Firestore



Firestore is a flexible and scalable non-relational database commonly used for mobile, web, and server development.



It can process NoSQL queries to retrieve data storage in the form of **documents**.



Documents are stored in **collections**.

Firestore features



Firestore



Queries can be filtered and sorted in Firestore, and are indexed by default.



Query performance is dependent on the result set instead of the dataset.

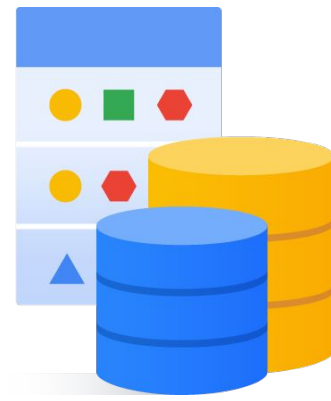


Firestore syncs data across all connected devices, but it's also optimized for simple, one-time queries.



Firestore uses data caching to enable an application to write, read, monitor, and query data even while a device is offline.

Spanner



A SQL-based, fully managed, scalable relational database.



It's the same database that powers Google's mission-critical apps and services.



Has high availability.

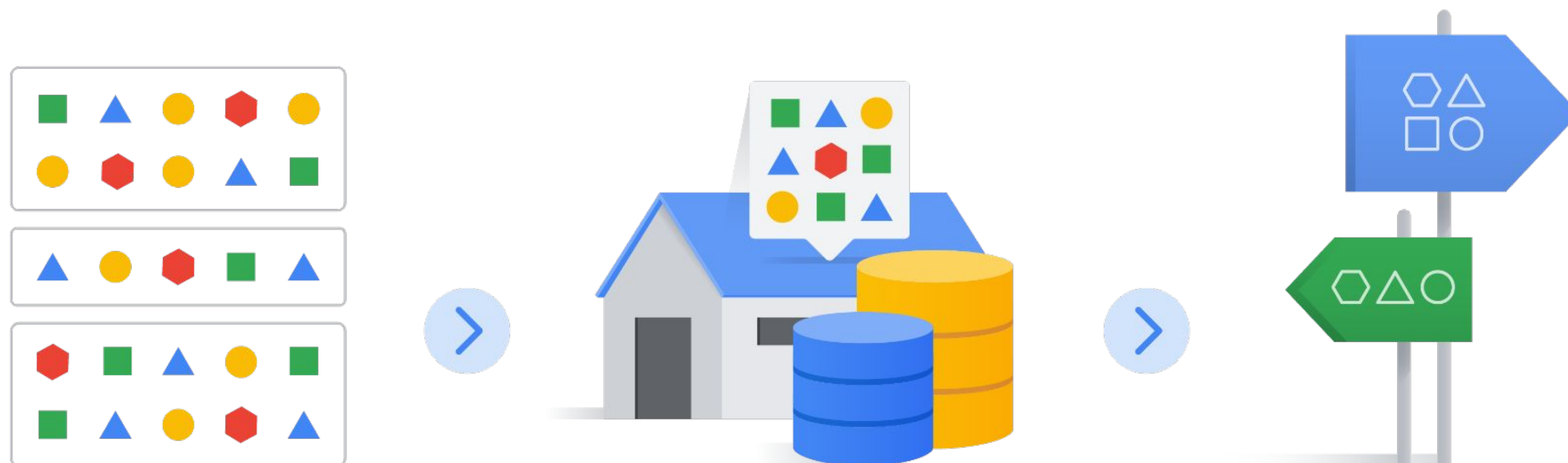


Is globally consistent.



Is capable of performing tens of thousands of read/write operations per second.

BigQuery



BigQuery is a fully-managed data warehouse.

BigQuery



BigQuery



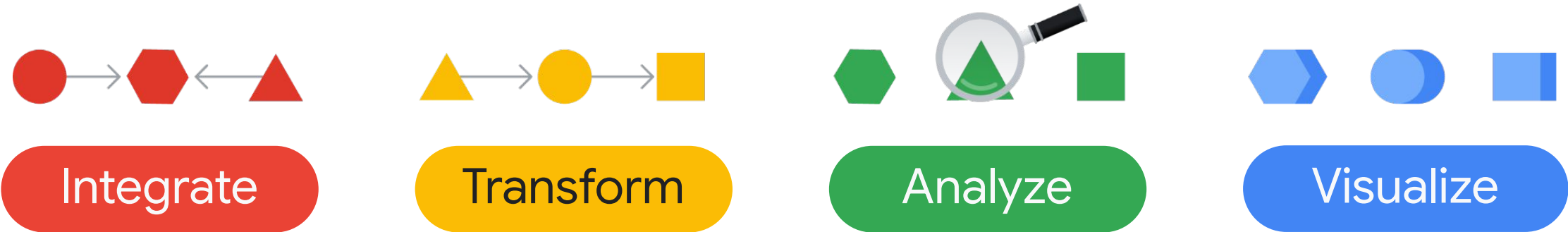
Provides a data warehousing backend for modern business intelligence solutions.



Helps guide management decisions.

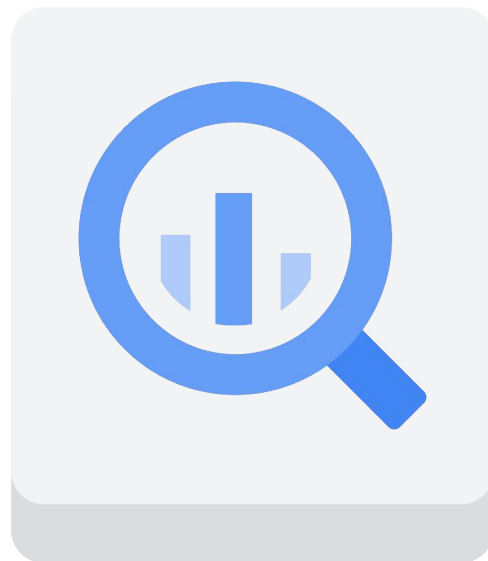


Manages the technical aspects of storing structured data.



You can integrate, transform, analyze, and visualize your data by using Google and third-party tools.

BigQuery features



BigQuery



Excels at storing and analyzing massive amounts of event and sensor data generated by IoT applications.



Optimized for 'write-once, read-many' data patterns.



Separates storage and compute costs. You only pay for query execution, not for data at rest.

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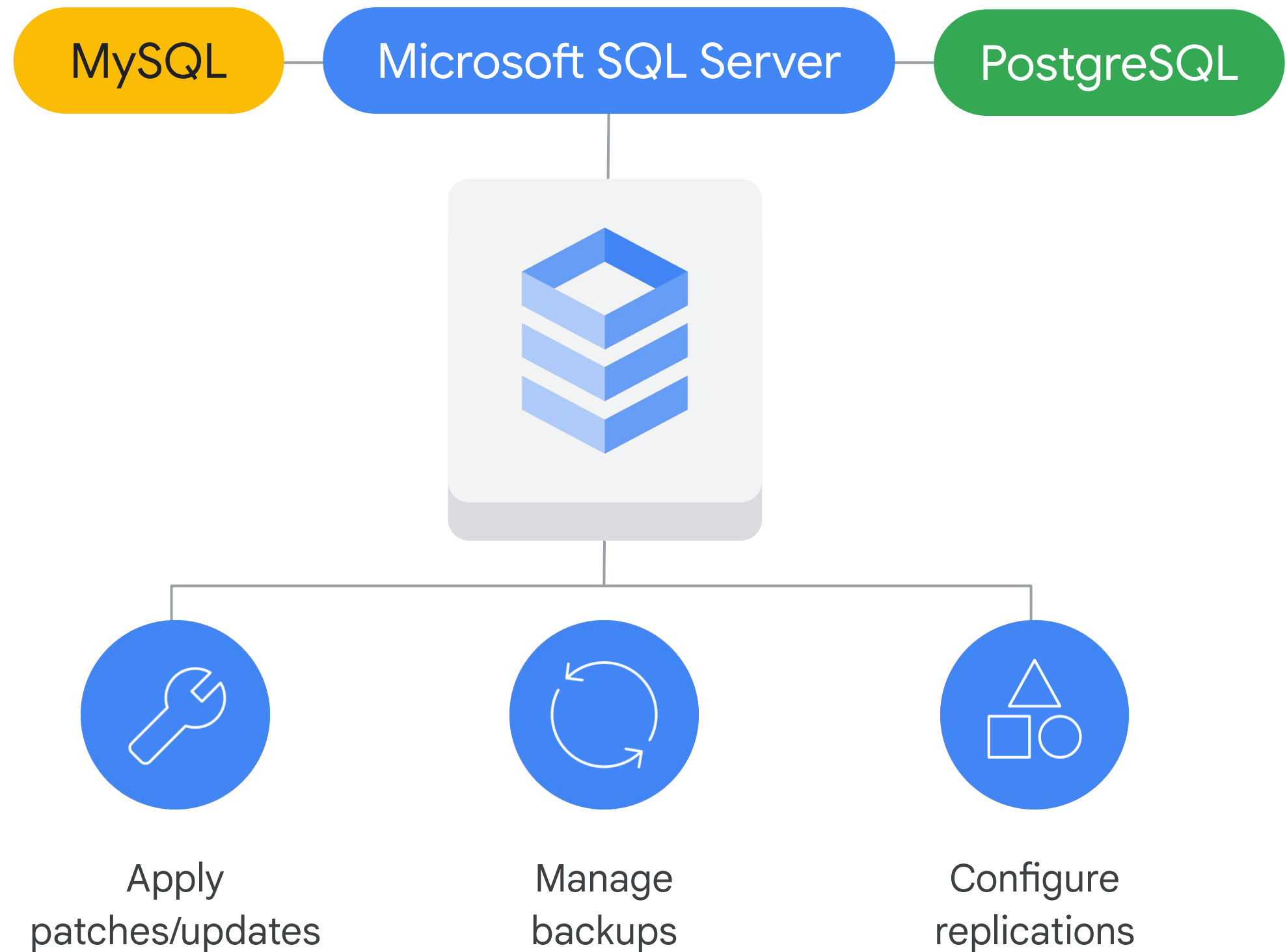
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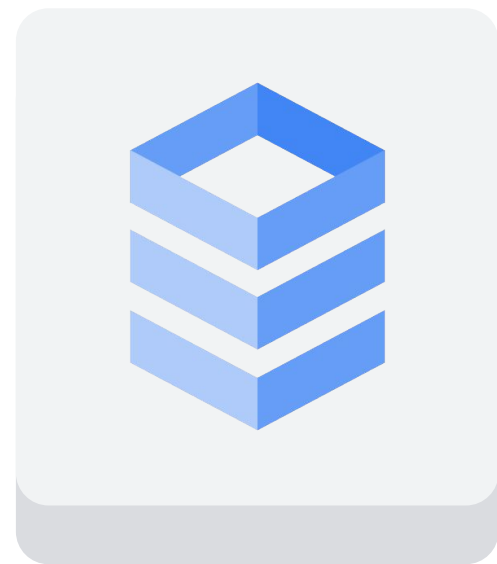


Cloud SQL

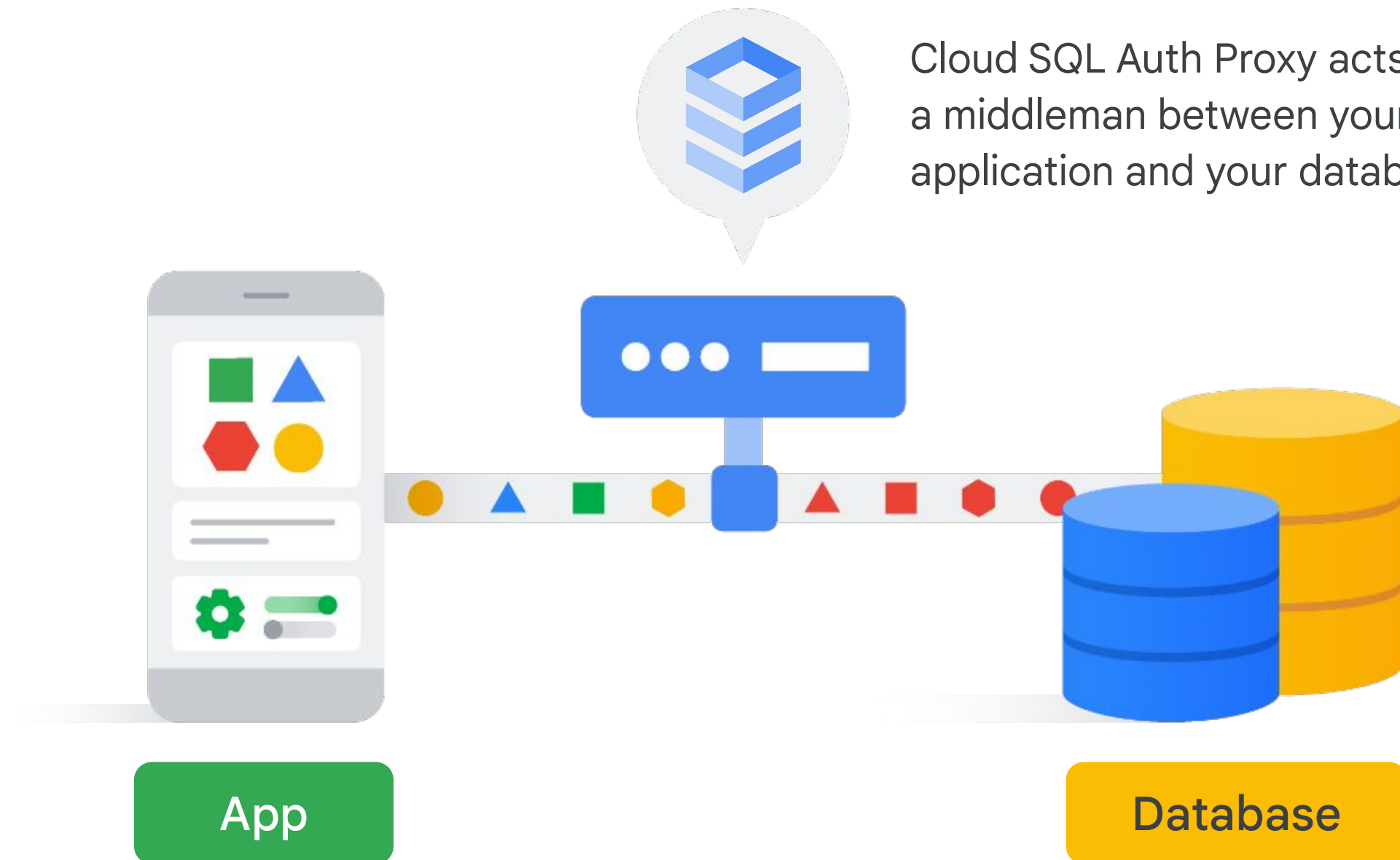
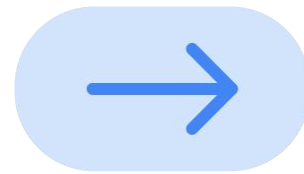
A fully managed database service that simplifies the process of setting up, maintaining, and scaling **relational databases**.



Cloud SQL Auth Proxy

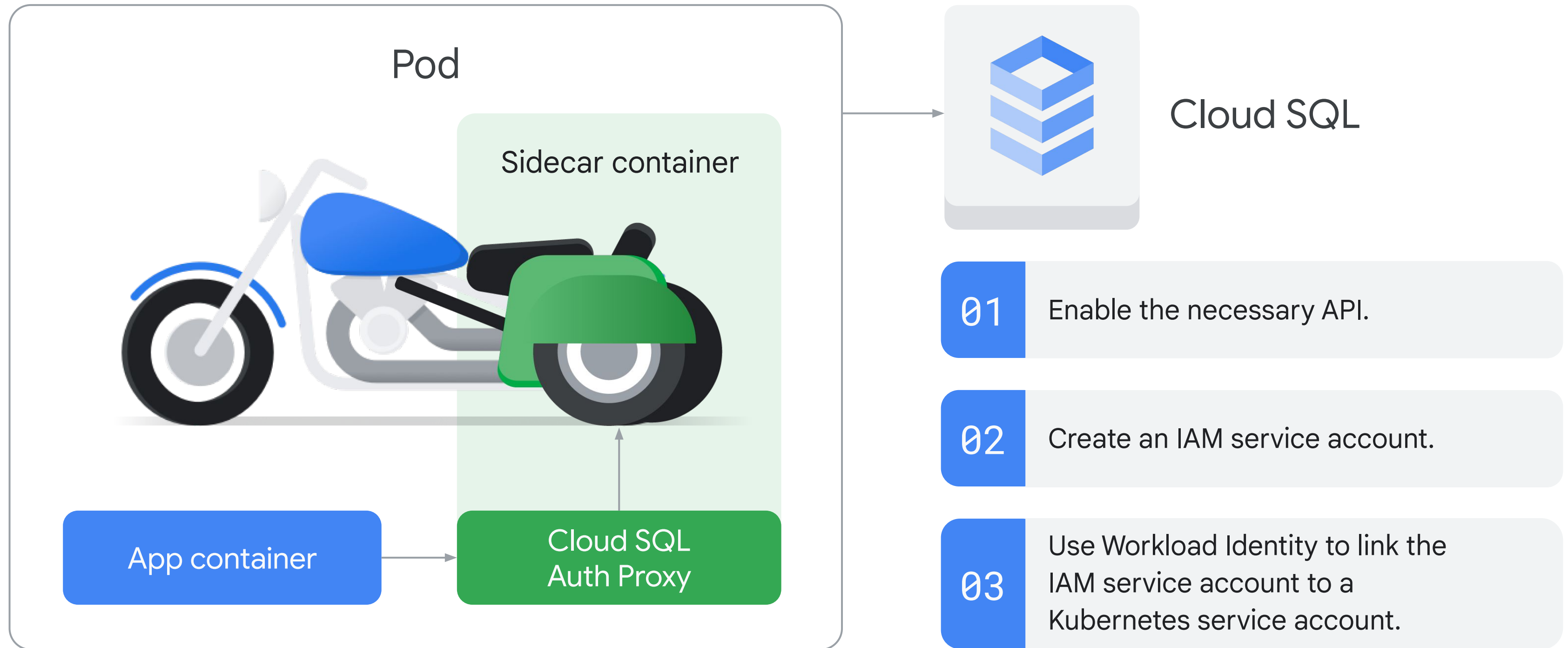


Streamlines and secures the connection to your Cloud SQL database instances.



Cloud SQL Auth Proxy acts as a middleman between your application and your database.

All Pods can reliably access the database

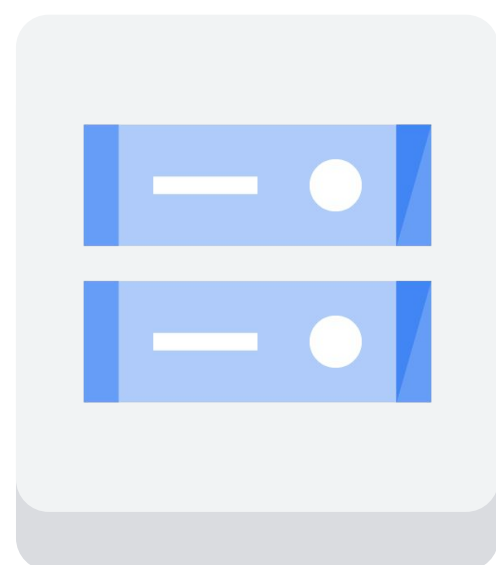


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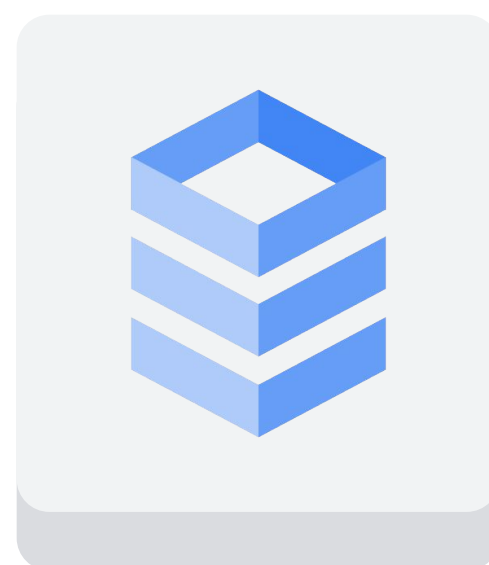
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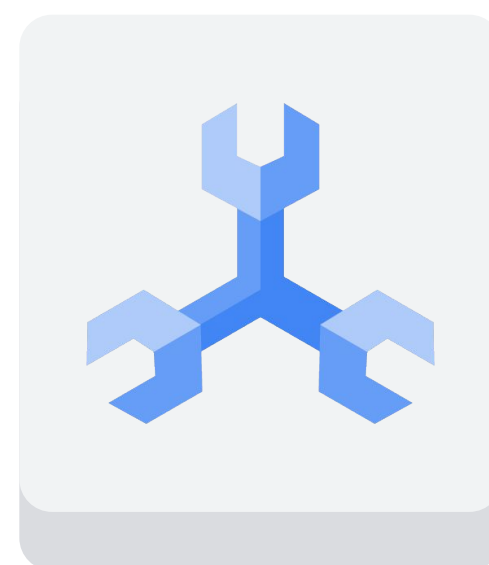
Choosing a storage option



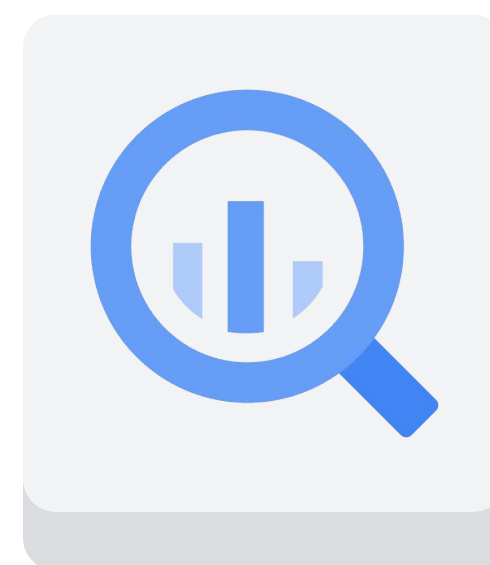
Cloud Storage



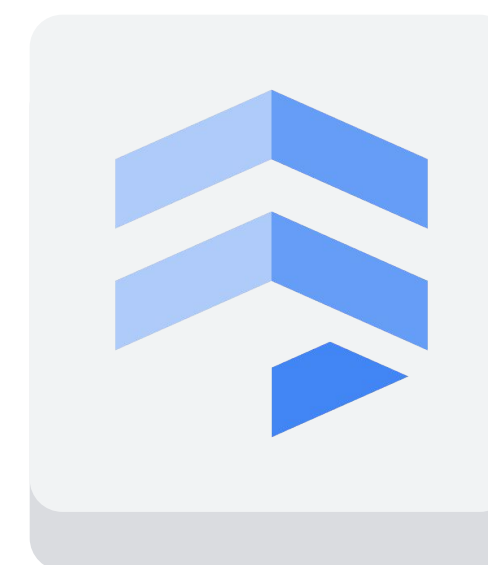
Cloud SQL



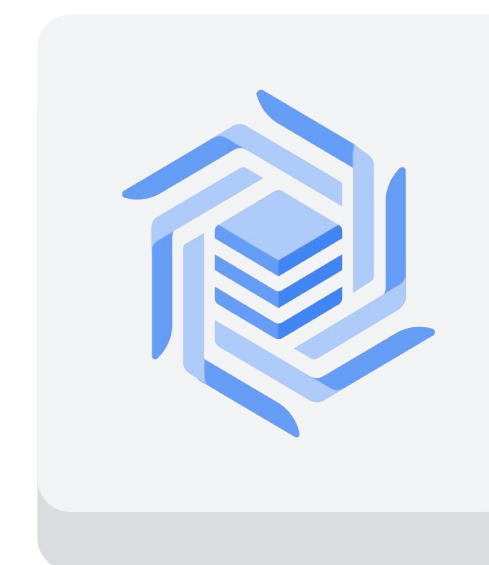
Spanner



BigQuery



Firestore



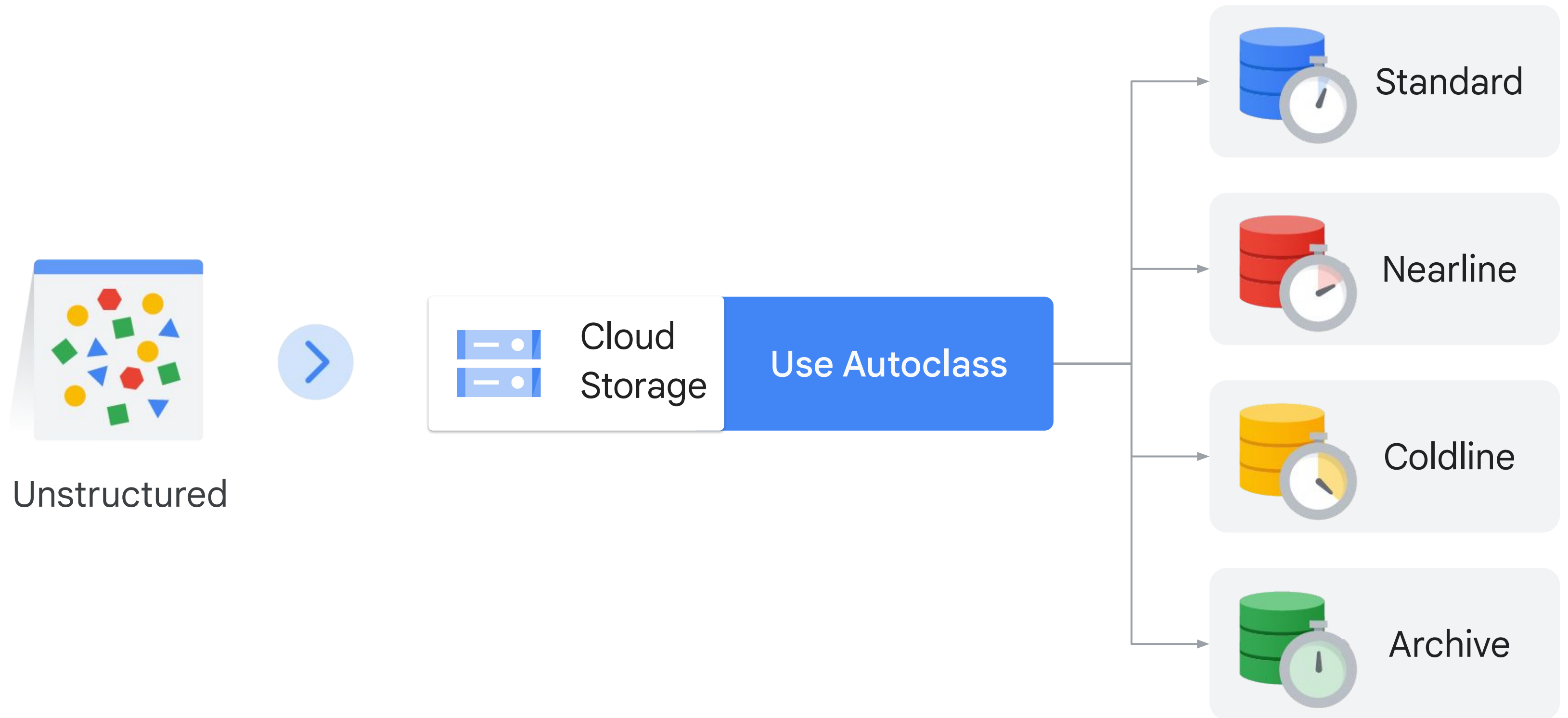
Bigtable

Data type

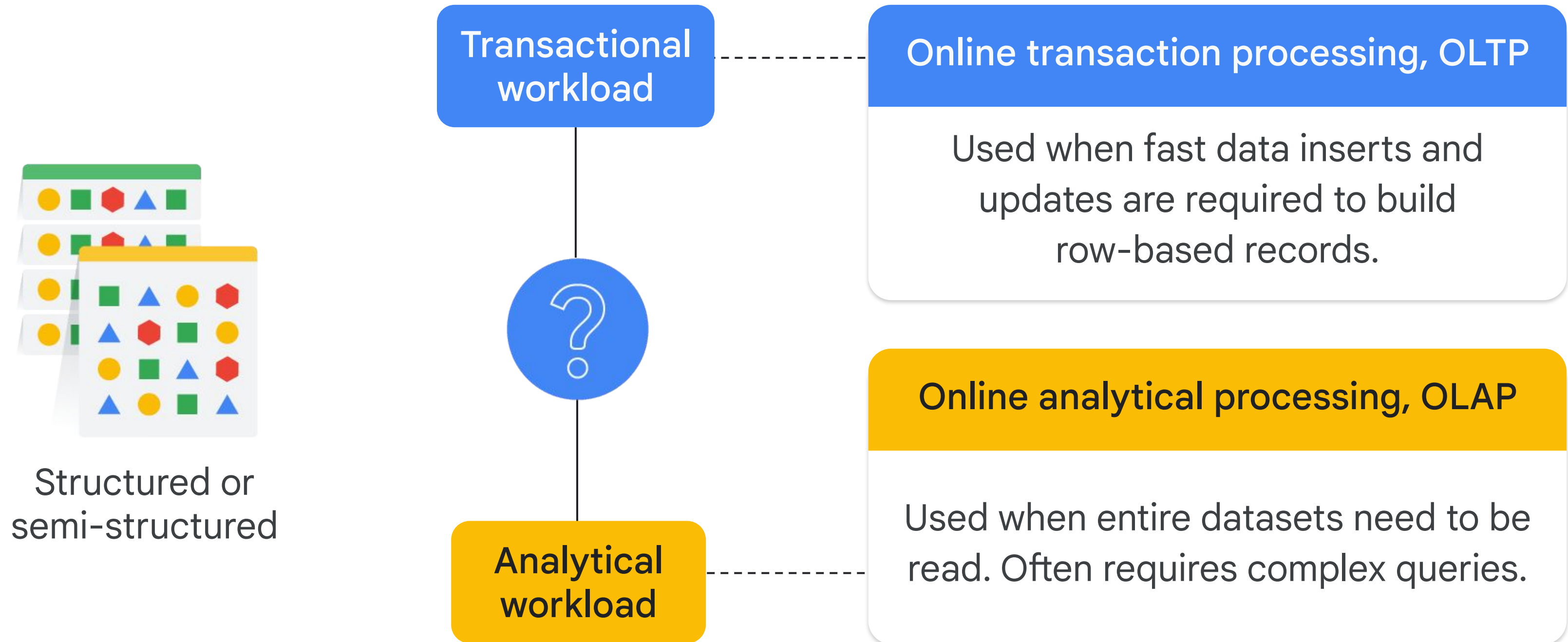


Business need

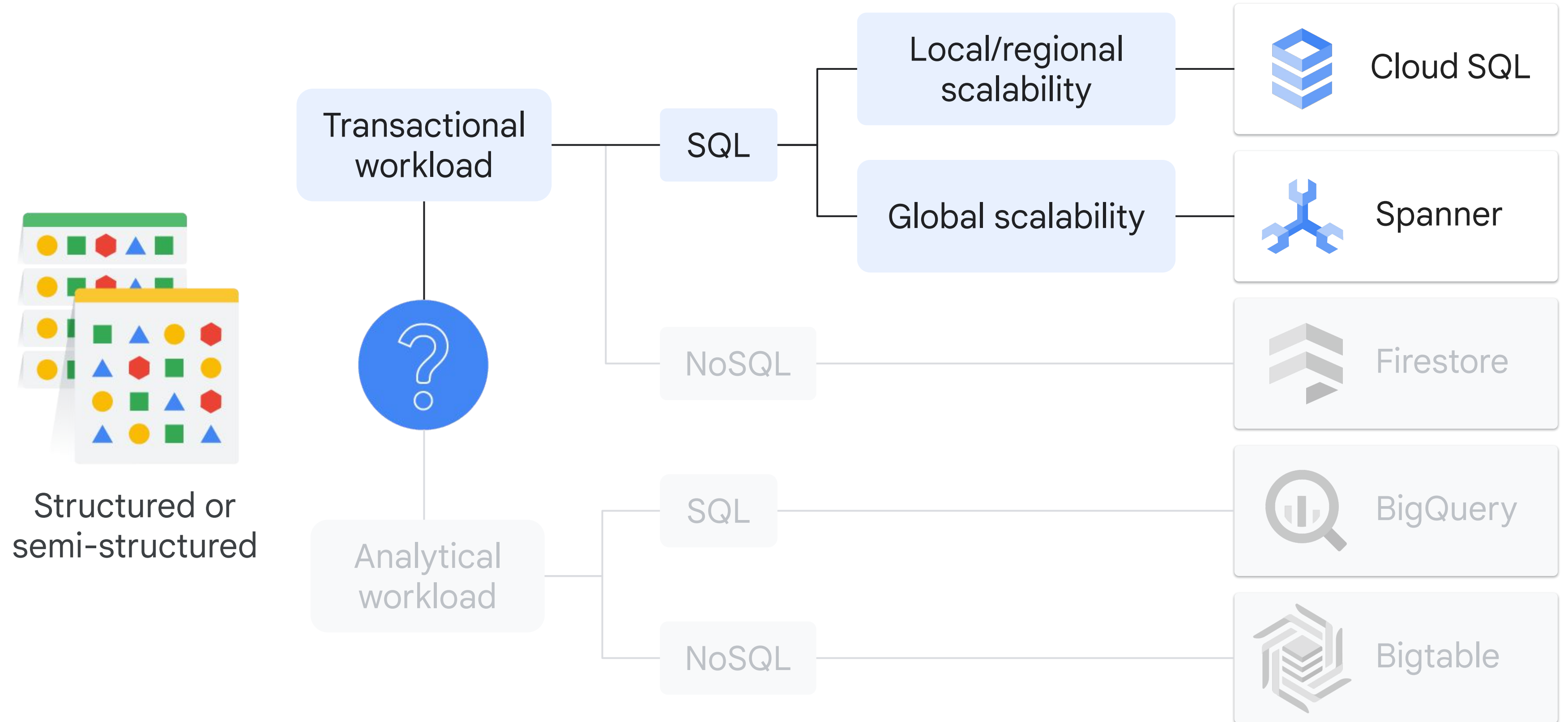
Use Cloud Storage for unstructured data



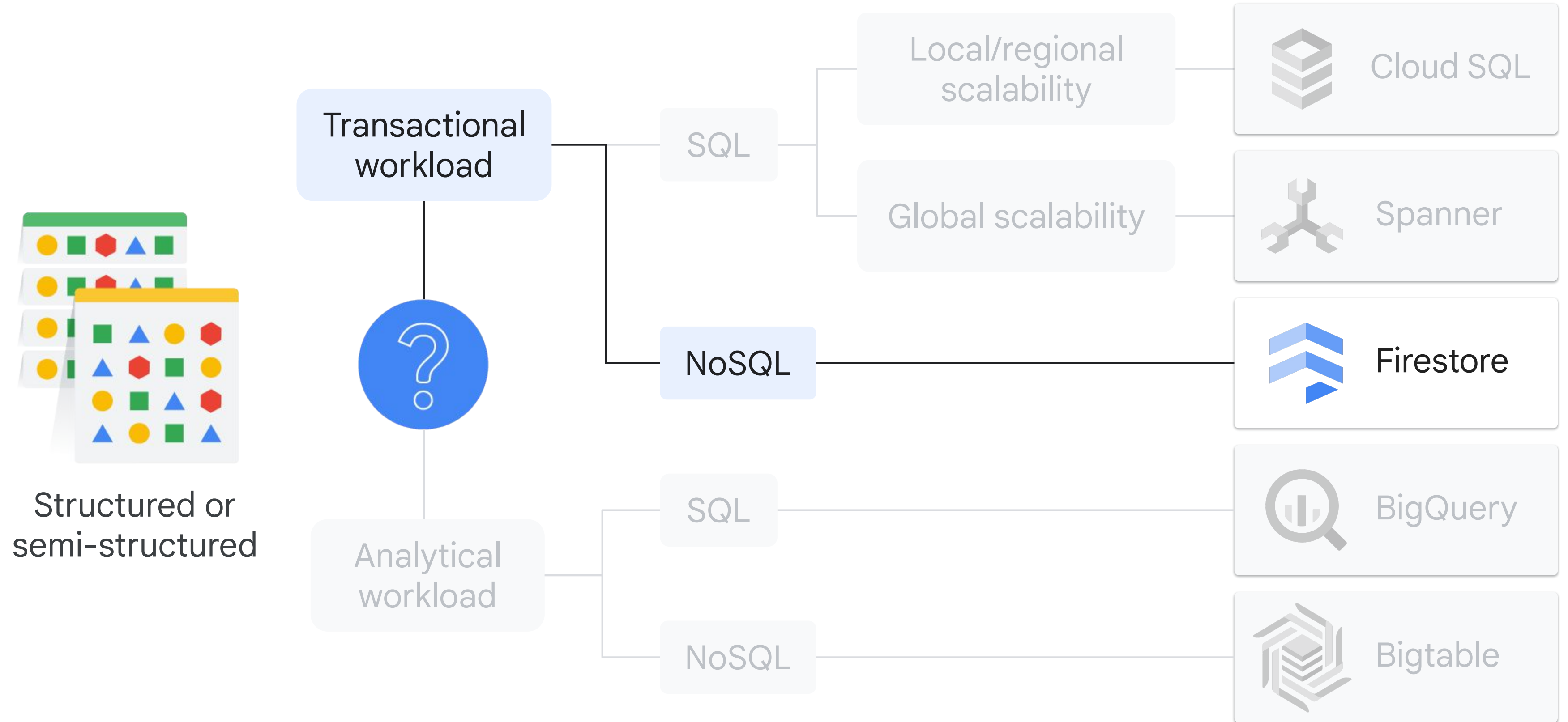
Structured or semi-structured data



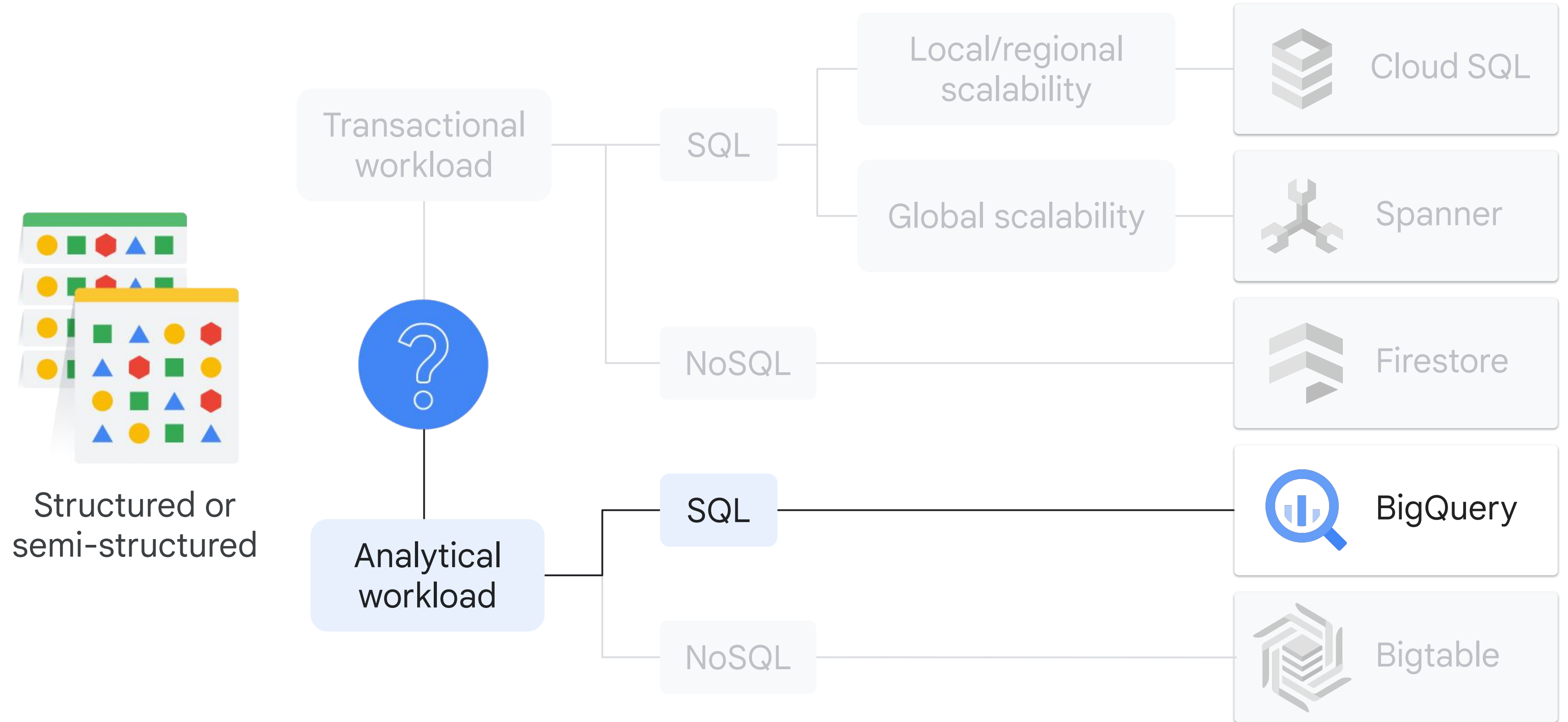
Transactional workloads accessed with SQL



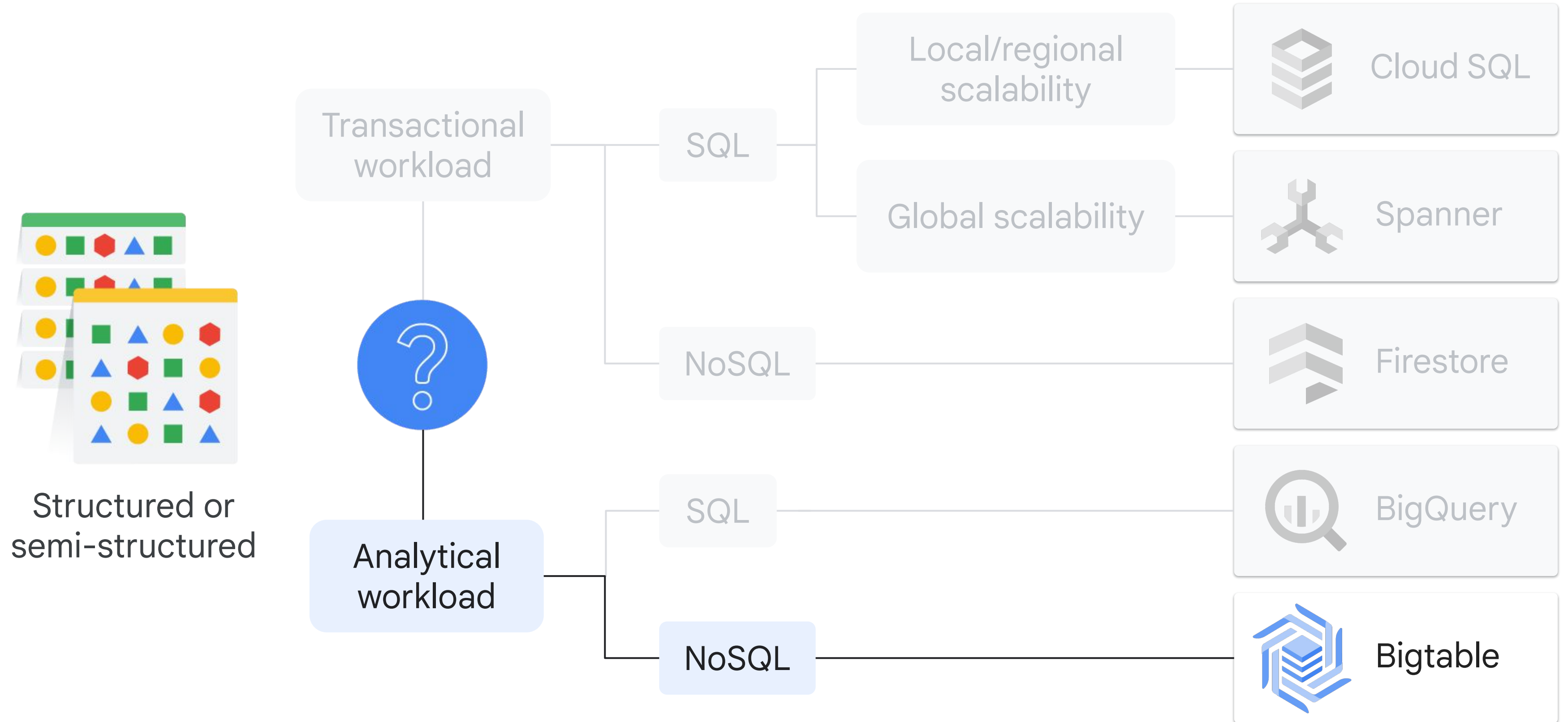
Transactional workloads accessed without SQL



Analytical workloads that require SQL commands

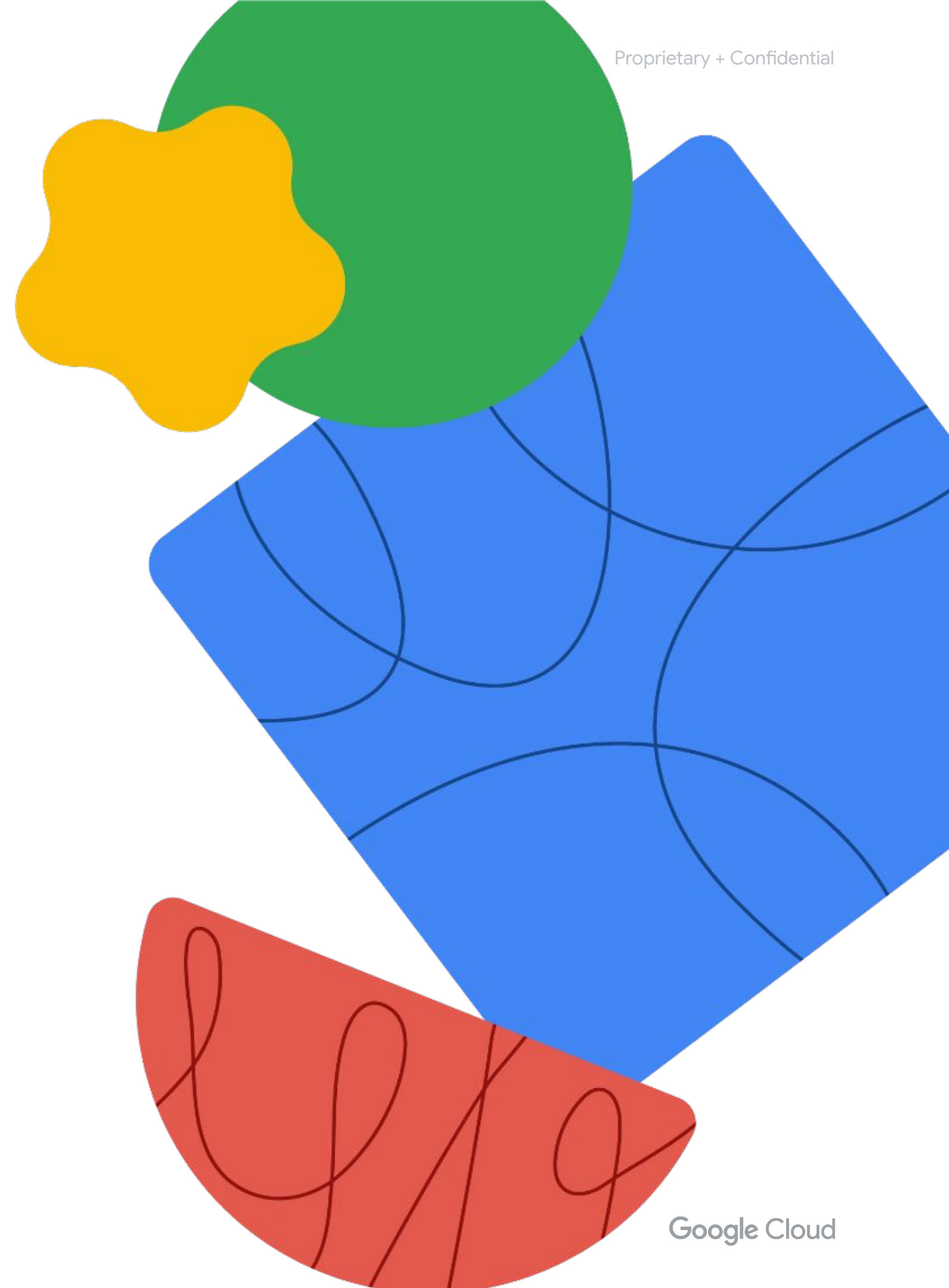


Analytical workloads that don't require SQL commands



Quiz questions

Let's pause for a quick check in.



Quiz | Question 1

Question

How do you enable a GKE Pod to securely access a Google Cloud storage service?

- A. Give the application the credentials of a Google IAM account with the necessary permissions to access the Google Cloud service.
- B. Deploy a proxy container with the service's credentials in the same Pod as your application.
- C. Grant the cluster's Kubernetes Service Account the necessary permissions to access the Google Cloud service using Workload Identity.
- D. Add a DNS SRV record for the managed service to the GKE cluster's DNS server.

Quiz | Question 1

Answer

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Quiz | Question 2

Question

You need a cloud storage solution for archiving business data to comply with regulations. The archives need to be kept for 5 years, audited quarterly, and rarely accessed otherwise. What solution best fits these needs?

- A. Cloud Storage - Standard
- B. Cloud Storage - Coldline
- C. Cloud Storage - Archive
- D. Cloud Storage - Nearline

Quiz | Question 2

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Quiz | Question 3

Question

You're building an online multiplayer game that needs to propagate a consistent global game state across a massive amount of concurrent sessions. Which Google-managed storage solution should you use?

- A. Cloud Bigtable
- B. Cloud SQL
- C. Cloud Storage
- D. Firestore

Quiz | Question 3

Answer

You're building an online multiplayer game that needs to propagate a consistent global game state across a massive amount of concurrent sessions. Which Google-managed storage solution should you use?

- A. Bigtable
- B. Cloud SQL
- C. Cloud Storage
- D. Firestore



Quiz | Question 4

Question

What is the purpose of SQL Auth Proxy?

- A. It is a tool that helps manage service accounts in Kubernetes.
- B. It allows applications running in Kubernetes to connect to Cloud SQL databases.
- C. It is a way to connect to databases that are hosted on Google Cloud.
- D. It is a way to manage access to databases in Google Cloud.

Quiz | Question 4

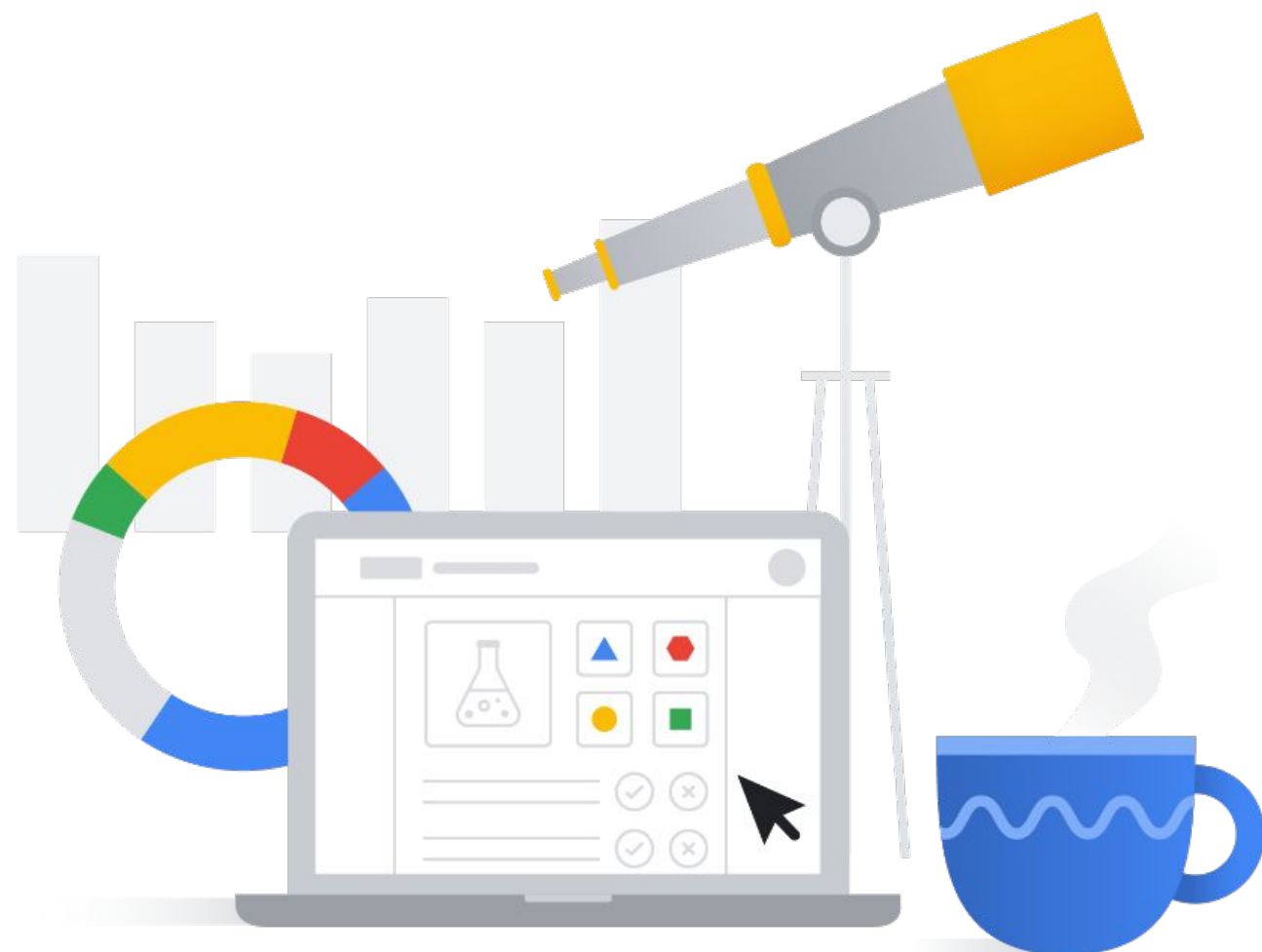
Answer

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Lab: Using Cloud SQL with Google Kubernetes Engine



01

Create credentials and Kubernetes Secrets for application authentication.

02

Configure Workload Identity.

03

Configure a Deployment with a Wordpress image to use SQL Proxy.

04

Install SQL Proxy as a sidecar container and use it to provide access to Cloud SQL.